

Forecasting for adaptive reservoir management

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**Agència Catalana
de l'Aigua**



Hydrological Stress

- Prolonged droughts reducing storage
- Intense rainfall causing flash floods
- Shifting seasonal precipitation patterns



Water Quality Risks

- Altered biogeochemical cycles
- Eutrophication & cyanobacteria blooms
- Dissolved oxygen depletion (hypoxia)

Increasing uncertainty for water supply operators & environmental managers

Why Seasonal Forecasting?



Weather
forecast

1–10 days



Sub-seasonal
forecast

2–6 weeks



Seasonal
forecast

1–7 months



Climate
projection

years–decades

← shorter · prediction horizon · longer →

The seasonal window (1–7 months) aligns with reservoir management decisions, drought contingency plans, irrigation scheduling, treatment optimization, and ecological flow management.

Modelling Approach



CLIMATE FORCING

- Seasonal climate forecasts (SEAS5)
- Multiple ensemble members



HYDROLOGICAL MODEL

- Catchment rainfall-runoff
- Storage & volume dynamics



LIMNOLOGICAL MODEL

- Thermal model (process-based)
- Ecology (ML)

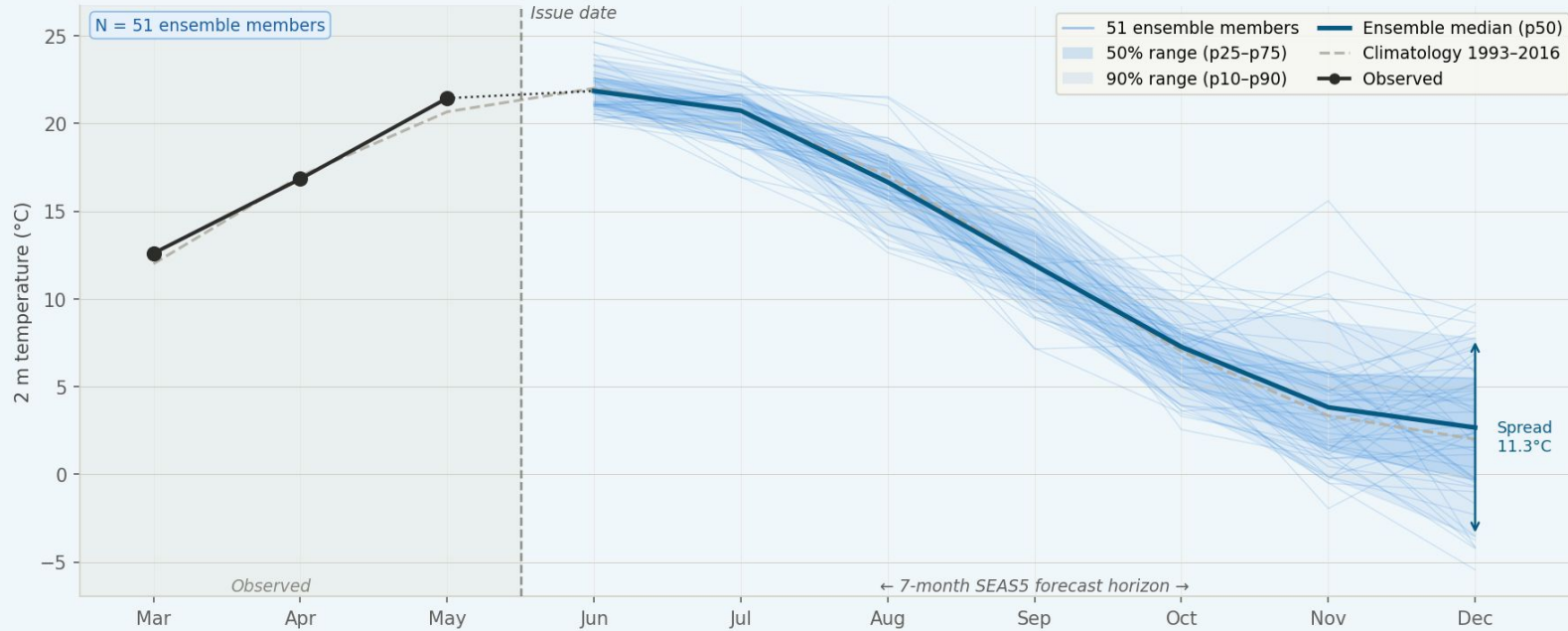


OUTPUT: Prediction of water quantity & quality: 1 to 7-month horizon

How does Seasonal Forecasting work?

SEAS5 seasonal forecast — 2 m temperature

Ter catchment · issued June · ECMWF SEAS5 (51-member ensemble)



1–7

months

Forecast horizon

51

members

Case Studies: The Ter River System



Ter Catchment — Catalonia

Catchment area	~3,010 km ²
Mean annual rainfall	~780 mm/yr
Climate type	Mediterranean
River length	208 km (Pyrenees → sea)

Water uses downstream



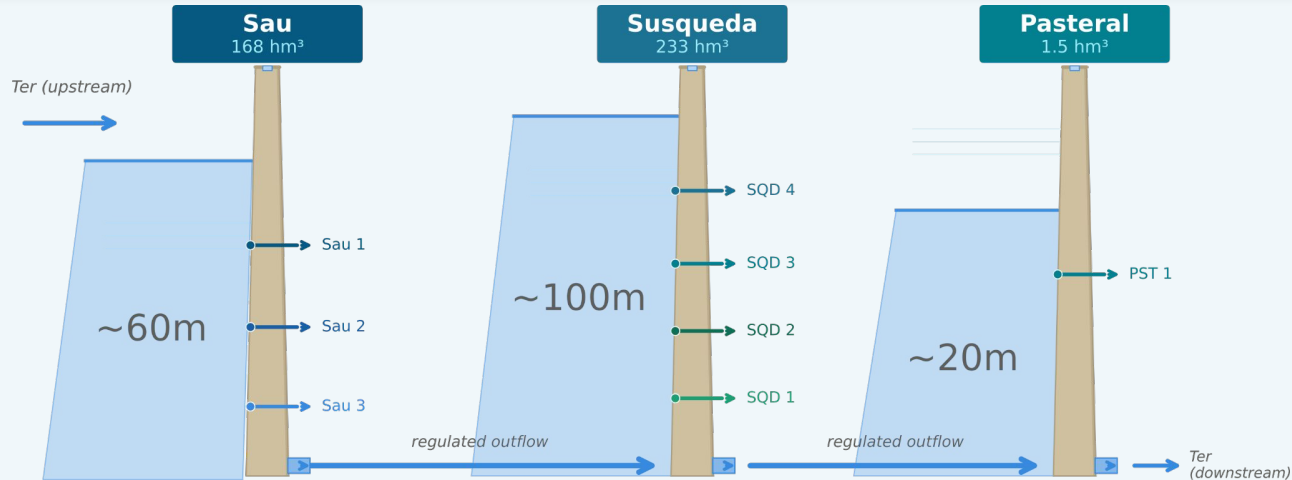
Barcelona metro area
(>3 M inhabitants)



Girona city
(>100k inhabitants)



Baix Ter plains
(~30,000 ha)



Case Studies: The Ter River System

Sentinel 2 data



Target Prediction Variables

WATER QUANTITY

Inflows

River inputs from catchment

Storage levels

Daily to monthly reservoir volume

Reservoir volume

Total water availability

WATER QUALITY

Temperature

Thermal stratification profiles

Nutrients (N, P)

Eutrophication risk indicators

Dissolved oxygen

Hypoxia risk in hypolimnion

Chlorophyll

Algal biomass & bloom potential

Adaptive Management Under Climate Change

Traditional reactive management is no longer sufficient as climate extremes intensify. Seasonal forecasting enables a shift from reactive to anticipatory management.

Reactive (today)

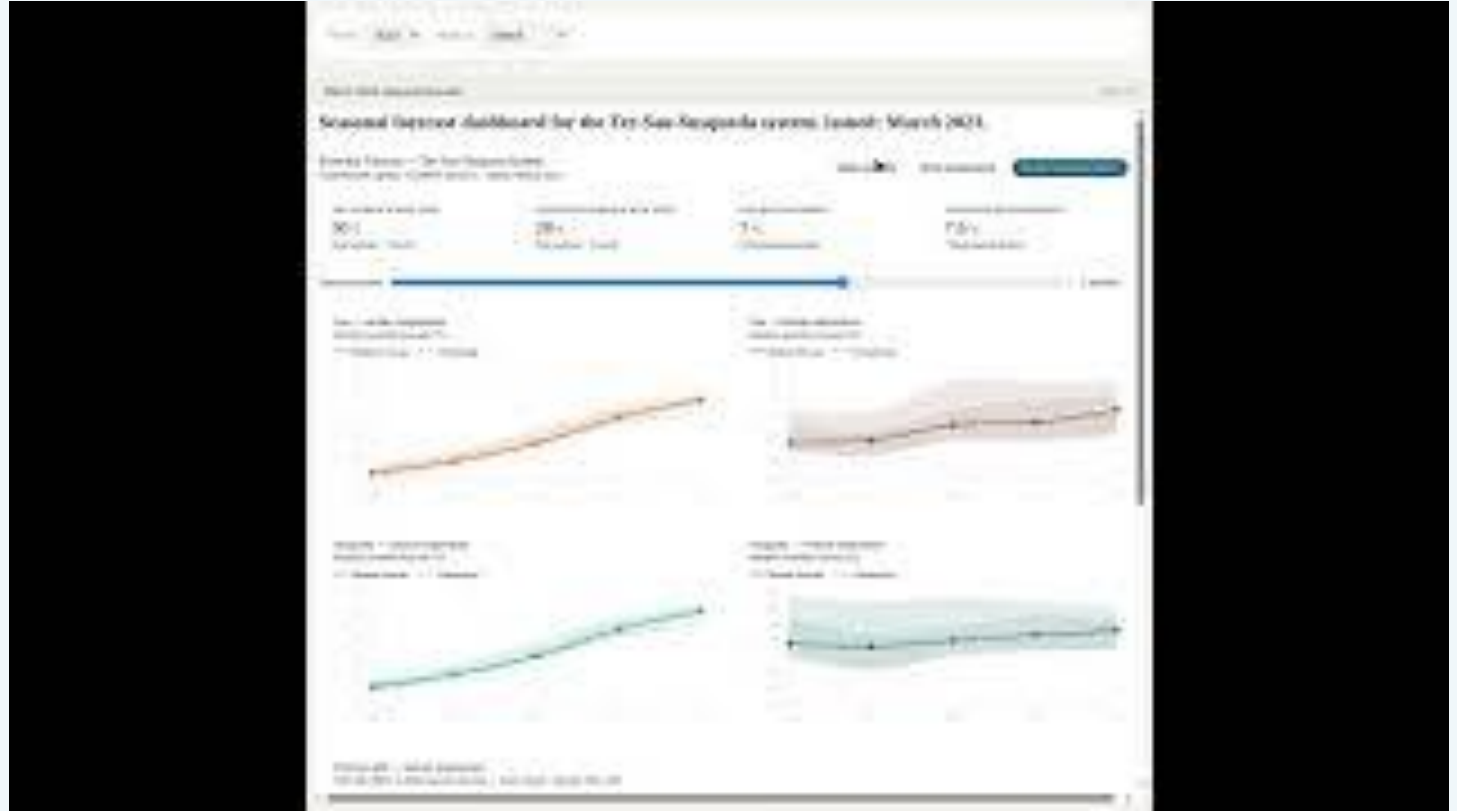
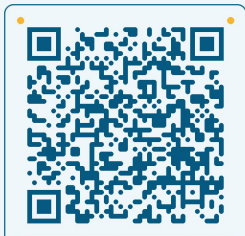
- Respond after drought onset
- React to quality alerts
- Conservative reserves always maintained
- Uniform treatment protocols year-round



Anticipatory (with forecasting)

- Pre-position resources 1–6 months ahead
- Proactive ecological interventions
- Optimized storage allocation
- Adaptive treatment strategy planning

Tool for the Ter system



https://nereidaca.github.io/forecasting_tool/

Limitations

Probabilistic nature of the climate data and uncertainty

Low predictability outside the tropics

Necessity of looking for windows of opportunities

Operational Relevance: Who Benefits?



Water Supply Operators

Anticipate scarcity 2–6 months ahead; trigger demand management & inter-basin transfers



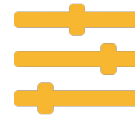
Environmental Managers

Ecological flow prescriptions; early warning for hypoxia and bloom events



Treatment Plant Managers

Adapt treatment protocols to predicted quality; reduce chemical costs



Policy & Planning

Drought contingency activation; climate adaptation strategy evaluation

Conclusions

- ✓ Integrated hydrological-limnological forecasting framework is feasible to produce seasonal predictions of water quantity and quality
- ✓ Applied to the Ter catchment (Sau, Susqueda, Pasteral), a critical system supplying Barcelona, Girona, and Baix Ter agriculture
- ✓ Seasonal forecasts (1–7 months) capture variability patterns relevant to operational management decisions
- ✓ Early risk information for water scarcity and quality deterioration can be delivered months in advance

Funding

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Thank you